

Electricity Price Prediction with LSTM

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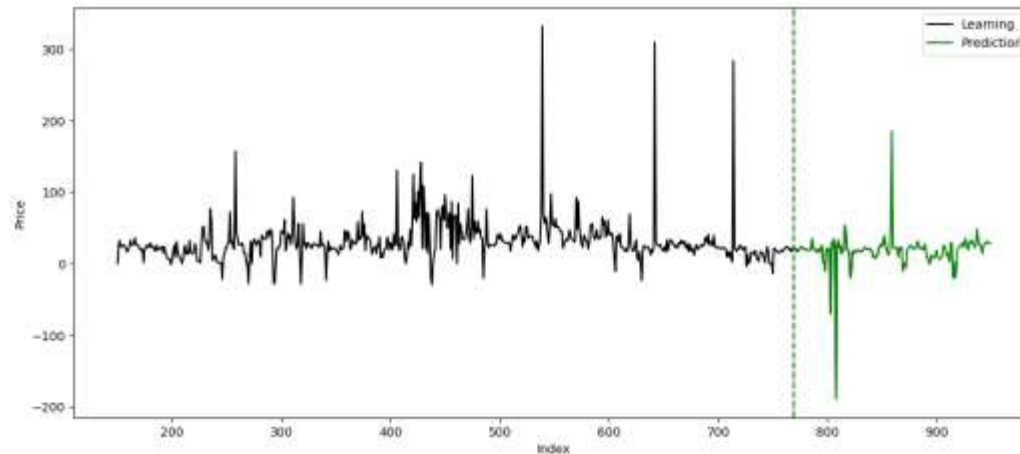
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Introduction

What is Electricity price prediction?

- Electricity price prediction is forecasting the cost of electricity over a future time period. Generally sorted as follows, Short-Term Forecasting (Minutes to days), Medium-Term Forecasting (usually months), Long-Term Forecasting (Sometimes years).



Sample instance of prediction



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Problem statement

- Electricity prices are highly volatile and influenced by various factors such as demand, supply, weather, and market dynamics, making accurate prediction is challenging and necessary.

Goal/Objective of this project.

- Want to predict hourly electricity price more accurately, decreasing the error values.

Impact of Electricity price prediction.

- **Energy Companies:** Optimizes generation scheduling and bidding strategies, enhancing operational efficiency.
- **Consumers:** Enables cost-effective energy usage and participation in demand response programs.

Sample input/output.

- Input usually contains various features that influence electricity price, represented as a vector of size K
Ex: (Feature 1, Feature 2 Feature K)
- Output is a single predicted value or a sequence of values representing future electricity prices
Ex: Price (in \$)

Feat 1	Feat 2	Feat 3	Feat 75	Target
17.01	7.67	14.67	313.0	24.68
67.11	-15.25	42.65	310.0	63.18
71.15	0.54	50.97	310.0	66.96
286.20	8.53	206.96	311.0	21.42

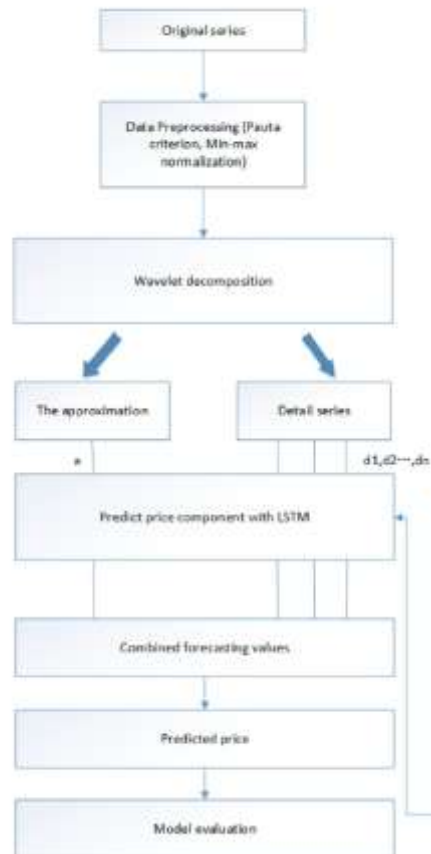
Sample Dataset



Existing Methodology

Method-1

[1] Electricity price prediction based on hybrid model of Adam optimized LSTM neural network and wavelet transform



1. **Wavelet Decomposition:** Transforming the price series into smoother sub-series for better pattern recognition.
2. **Adam-Optimized LSTM:** Training an LSTM network to capture sequential patterns and improve prediction accuracy using the Adam optimizer.
3. **Model Evaluation:** Using metrics like Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) to validate performance.

	BP	GRNN	LSTM	WT-Adam-LSTM
MAPE	31.2	31.2	18.42	7.56
MAE	10.1	10.1	5.53	2.77
RMSE	11.74	11.74	8.86	3.46

Method-2

[2] Hourly electricity price forecasting with NARMAX

This methodology of this **NARMAX model (Nonlinear AutoRegressive Moving Average model with eXogenous inputs)** for electricity price prediction involves:

- 1.Data Preprocessing:** Collecting and merging data from energy-related factors like electricity demand, generation, and weather, followed by averaging non-hourly data.
- 2.Peak Lag Identification:** Using autocorrelation analysis to identify key lagged terms from input variables that influence electricity prices (e.g., previous day or hourly lags).
- 3.NARMAX Modeling:** Building a predictive model by incorporating significant lag factors and refining the structure through iterative removal of redundant terms.

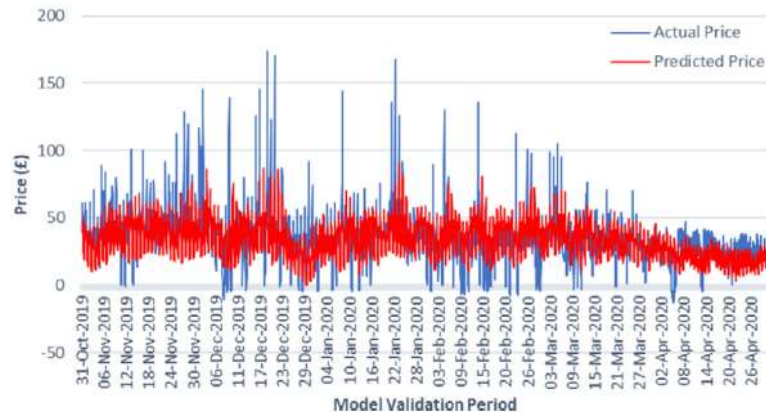


Fig. 16. Model validation for correlated lags SARIMAX model.

RMSE values for each model.

Model	RMSE
NARMAX	15.15
Correlated NARMAX	15.02
Correlated SARIMAX(3, 1, 3)(2, 0, 1, 24)	14.36
Refined correlated SARIMAX(3, 1, 3)(2, 0, 1, 24)	13.99

Method-3 (Base Paper)

- [3] Real-Time Energy Price Forecasting using BiLSTM-Autoencoder Deep Learning Model

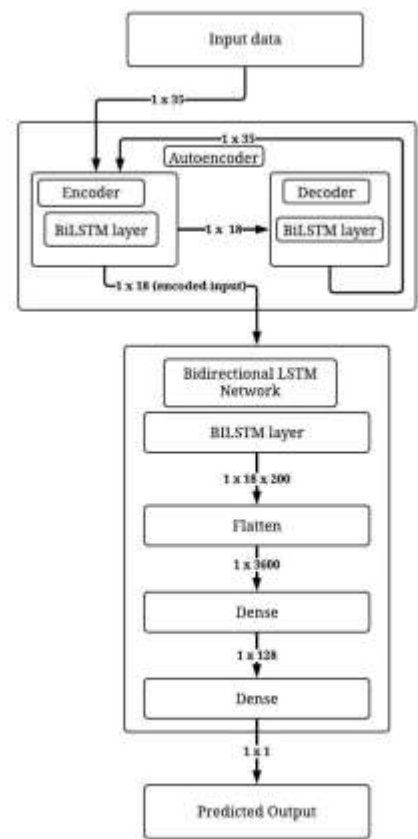


Fig. 3: Autoencoders with BiLSTM Architecture

Feature Selection:

- From 75 input features, significant ones are selected based on Pearson correlation analysis, narrowing it down to 35 relevant features (e.g., historical prices, demand, and generation).

Preprocessing for Model Training:

- Negative values in the price data are converted to positive integers.
- Data is normalized using logarithmic functions to facilitate model training.

Model	MAE	RMSE	MAPE (%)
Multiple Regression	0.0094	0.0287	13.2749
LSTM	0.0085	0.0285	11.9570
BiLSTM	0.0075	0.0285	10.5865
AE-BiLSTM	0.0057	0.0187	7.0819



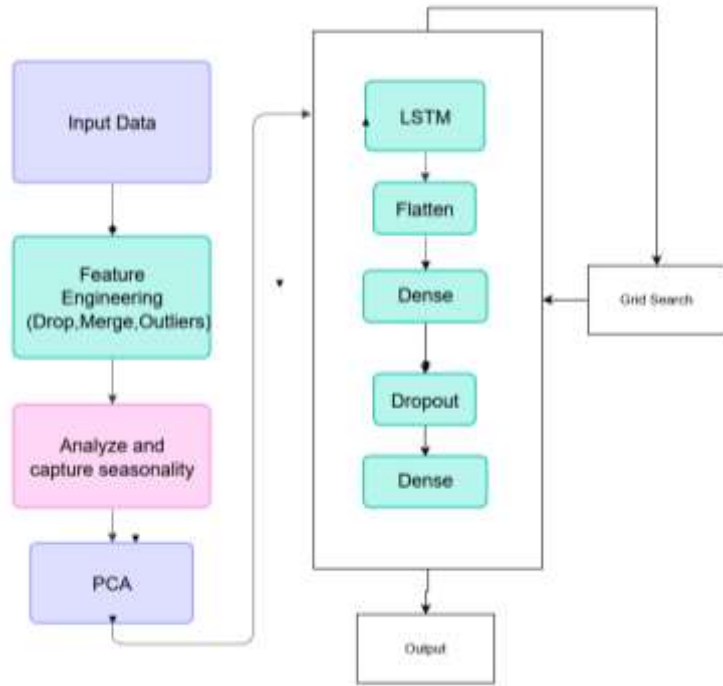
Gaps/Limitations of current work

- The current feature selection may overlook complex relationships or fail to capture all relevant predictors.
- The current models employed for electricity price prediction are highly complex
- Require extensive computational resources and prolonged training times.

Dataset Understanding

- Derived from the New York Independent System Operator (NYISO)
- Comprises 75 input variables, including historical electricity prices, load demand data from various sources.
- Contains 70128 tuples.

Our Approach



Implementation

- **Preprocessing:** Clean input data (drop, merge, handle outliers).
- **Feature Reduction:** Capture seasonality and apply PCA.
- **Modeling:** Train LSTM with Dense and Dropout layers.
- **Optimization:** Use Grid Search for tuning and generate output.

Feature understanding

- The dataset contains several lagged features, which may increase the model complexity without improving its performance.
- Many features exhibit high inter-correlation, potentially introducing multicollinearity into the model.

Feat 13	70127.0	21.823503	-3958.17	13.86	19.69	25.86	1886.7	35.438882
Feat 14	70127.0	27.30814	-577.55	16.97	22.58	29.515	2462.76	35.049515
Feat 15	70127.0	27.255612	-206.0	14.76	20.79	28.03	3906.41	56.34447
Feat 16	70127.0	31.069821	-1086.24	18.4525	24.28	32.7	1734.83	45.916948
Feat 17	70127.0	31.069746	-1086.24	18.45	24.28	32.7	1734.83	45.916968
Feat 18	70127.0	31.069853	-1086.24	18.455	24.28	32.7	1734.83	45.916939
Feat 19	70127.0	31.069849	-1086.24	18.455	24.28	32.7	1734.83	45.91694
Feat 20	70127.0	31.07067	-1086.24	18.455	24.28	32.7	1734.83	45.916975
Feat 21	70127.0	31.071138	-1086.24	18.455	24.28	32.7	1734.83	45.916958
Feat 22	70127.0	31.07125	-1086.24	18.455	24.28	32.7	1734.83	45.916932
Feat 23	70127.0	31.071242	-1086.24	18.455	24.28	32.7	1734.83	45.916934
Feat 24	70127.0	31.071231	-1086.24	18.455	24.28	32.7	1734.83	45.916938
Feat 25	70127.0	31.071267	-1086.24	18.455	24.28	32.7	1734.83	45.91693
Feat 26	70127.0	31.071255	-1086.24	18.455	24.28	32.7	1734.83	45.916933
Feat 27	70127.0	31.071181	-1086.24	18.455	24.28	32.7	1734.83	45.916948

Feat 66	
count	70128.0
mean	0.0
std	0.0
min	0.0
25%	0.0
50%	0.0
75%	0.0
max	0.0

Feature 66 info

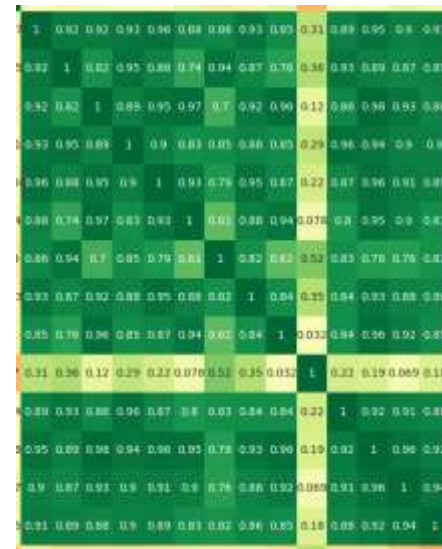
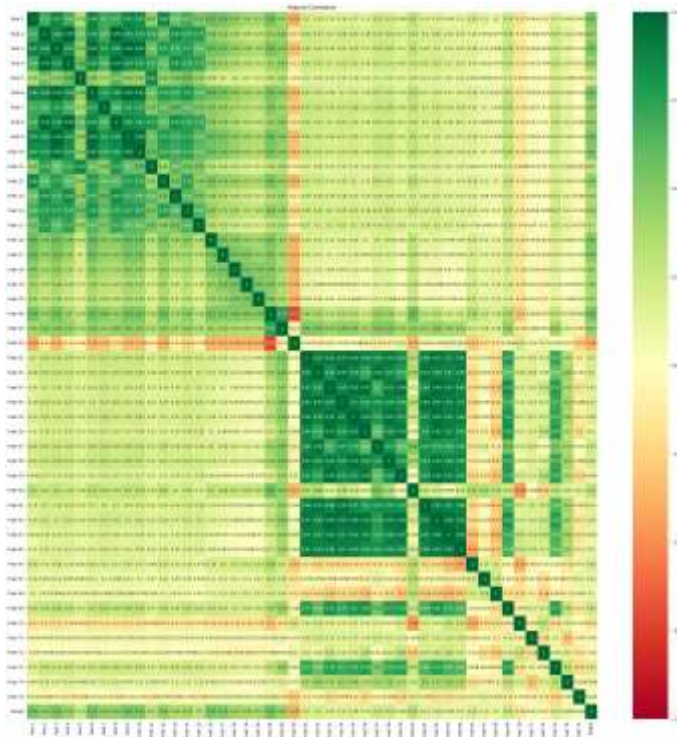
Identification of Lagged Features



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Preprocessing (Drop and Merge)

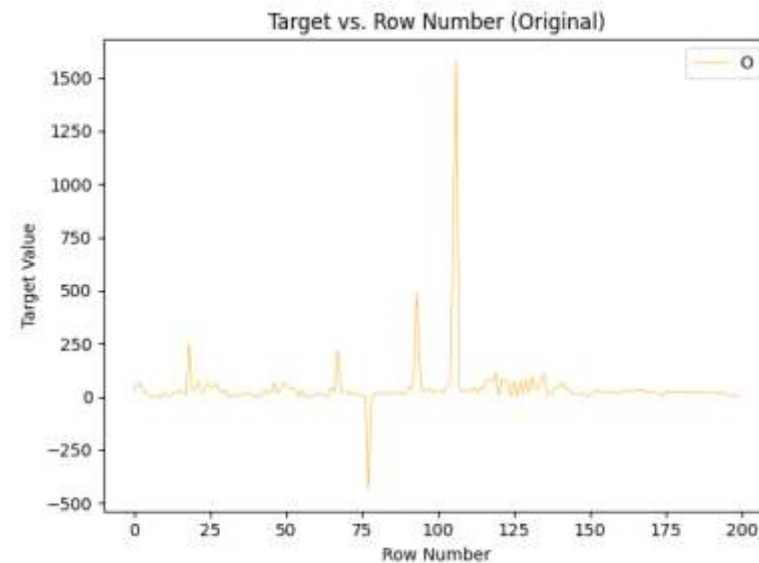
- We drop **Feat 66** and **most lagged features** except first 5 as it gave best accuracy
- We merge the features that have high correlation between themselves, helps in reducing dimensionality while retaining most relevant information
(We do it multiple steps so we can extract multiple combined features)



Zoomed in Cluster for Correlation

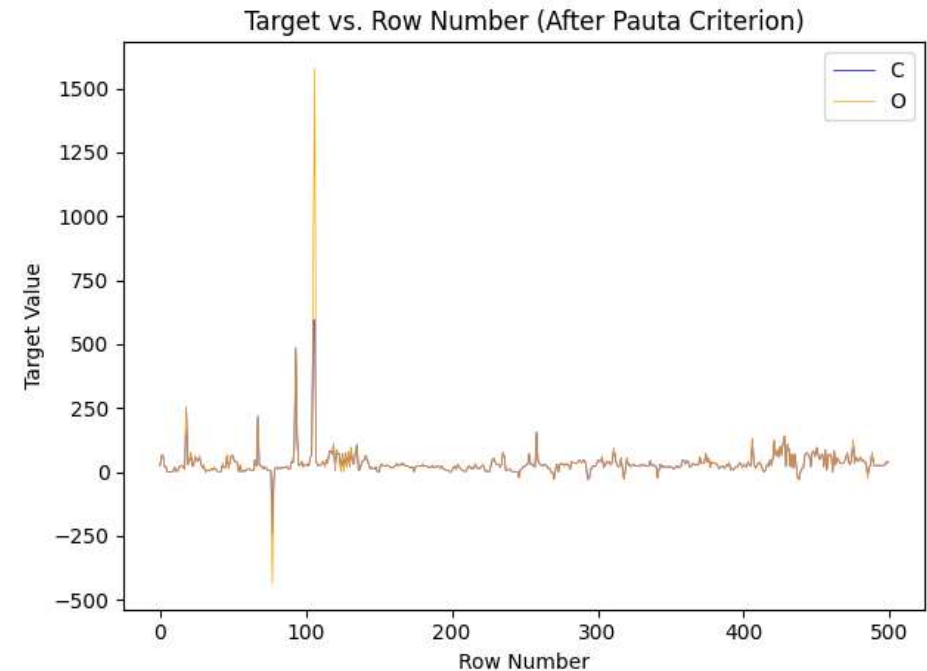
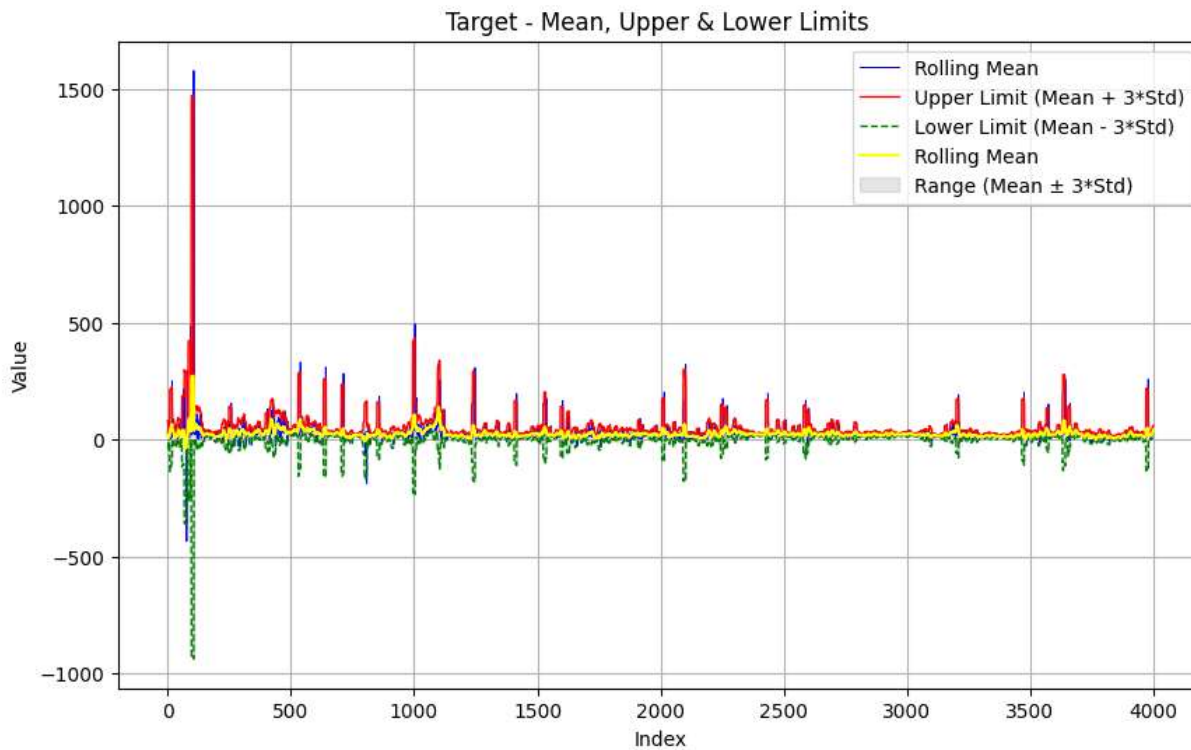
Outlier detection and handling

- Outliers are data points that deviate significantly from the rest of the dataset.
- Since our dataset has quality data, we only want to say a point is outlier if it is on an extreme end.
- Identifying and handling outliers is essential as they can **distort statistical analyses and negatively impact model performance**.



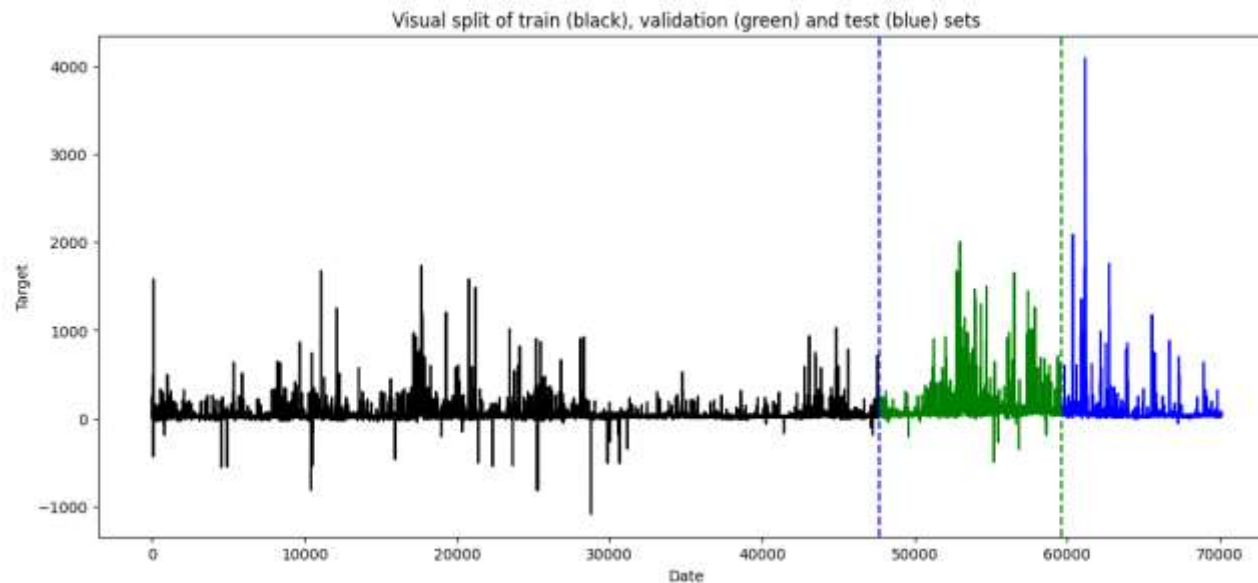
Outlier detection and handling

- We use the 3-sigma rule (Pauta criterion) to identify outliers
- We replace the identified outliers with mean of local **window (neighborhood)**
- **Windowing** helps us deal with seasonal highs and lows, making data less prone to be detected as outlier as datapoints in the window are used as comparison.
- For example if points are [150, 152, 148, 151, 149, 147, 600], then 600 is not an outlier.



PCA & Data Splits

- We applied PCA to reduce the dataset's dimensionality while retaining 95% of the variance.
- The process involves scaling the data (using MinMaxScaler) before performing PCA to ensure uniformity.(Makes training easier)
- We split the dataset into first 75% test , 10% validation, 15% test sets as follows

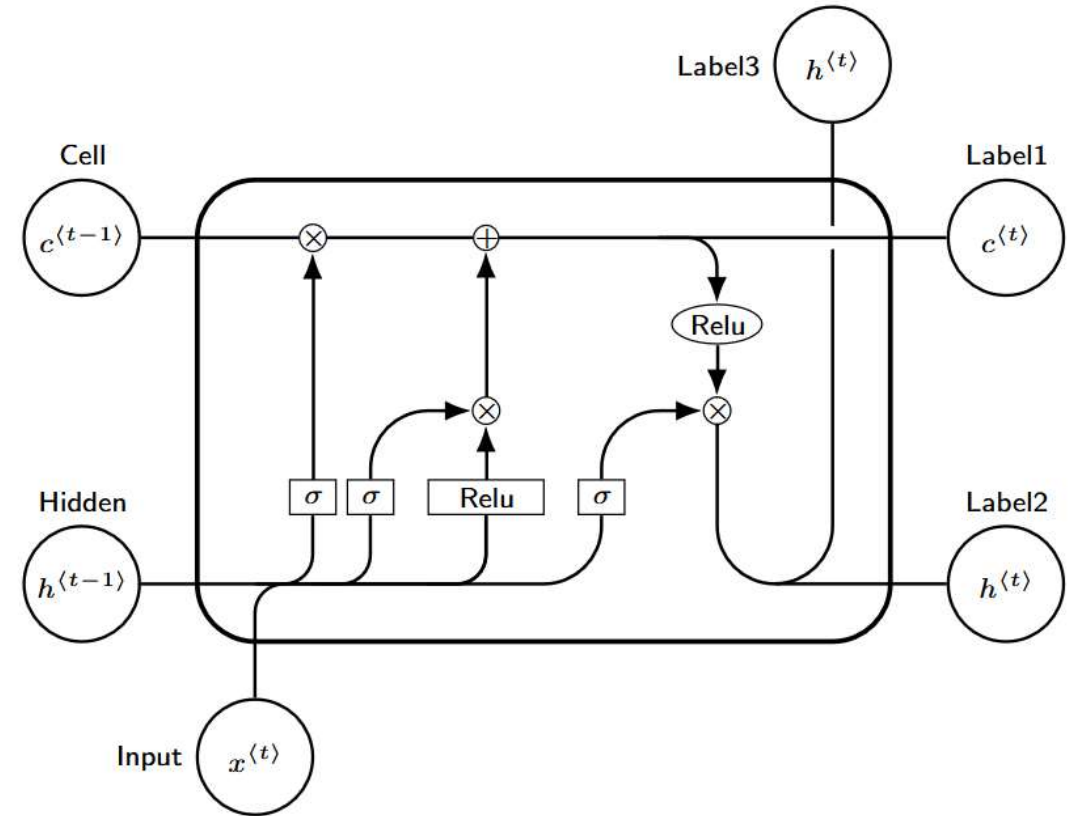


LSTM model

Architecture

A sequential LSTM model is defined with the following layers:

- **LSTM Layer:** Captures temporal dependencies with 128 units, using ReLU activation.
- **Flatten Layer:** Flattens the output from the LSTM layer.
- **Dense Layer:** Fully connected layer with 128 units and ReLU activation.
- **Dropout Layer:** Includes a 10% dropout rate to prevent overfitting.
- **Output Layer:** Final dense layer with a single unit for predictions.



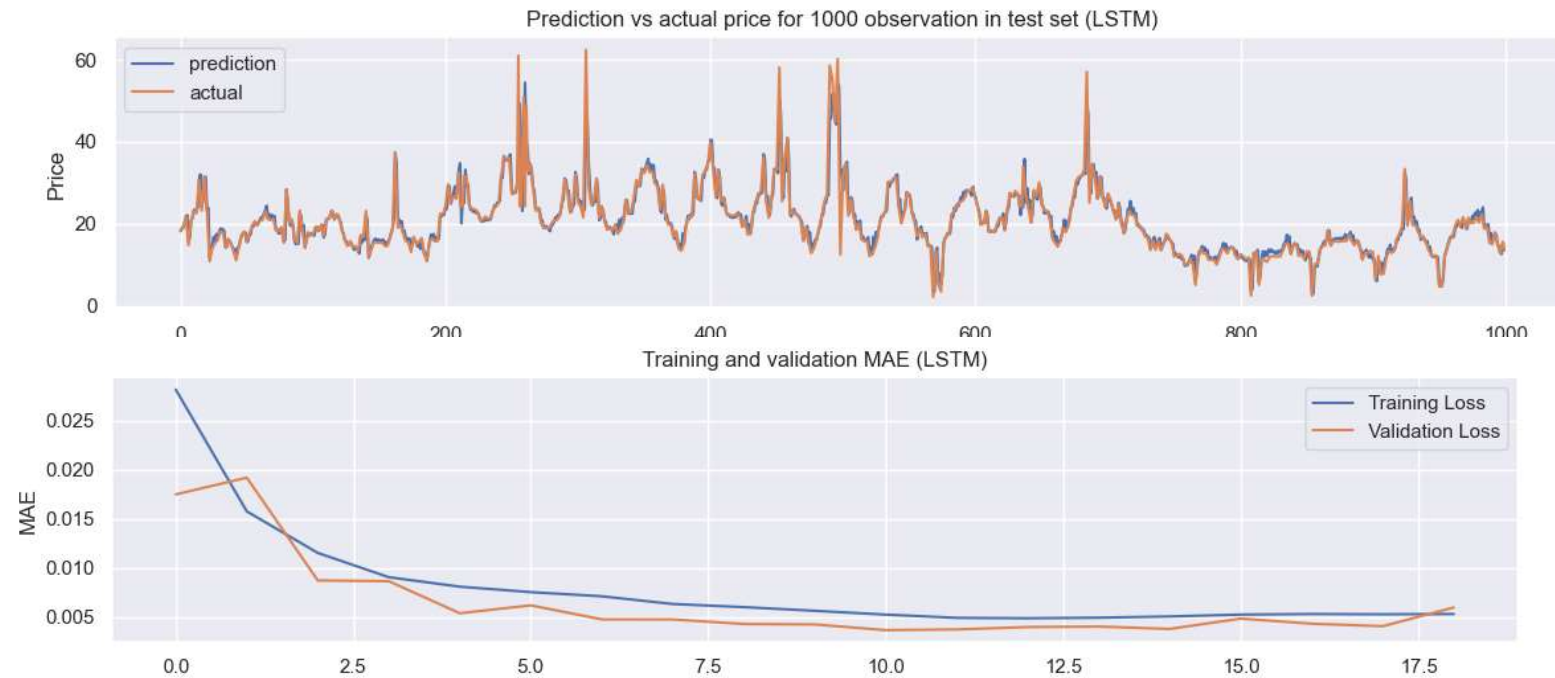
Results:

Error Metrics of our Model :

MAE: 1.356

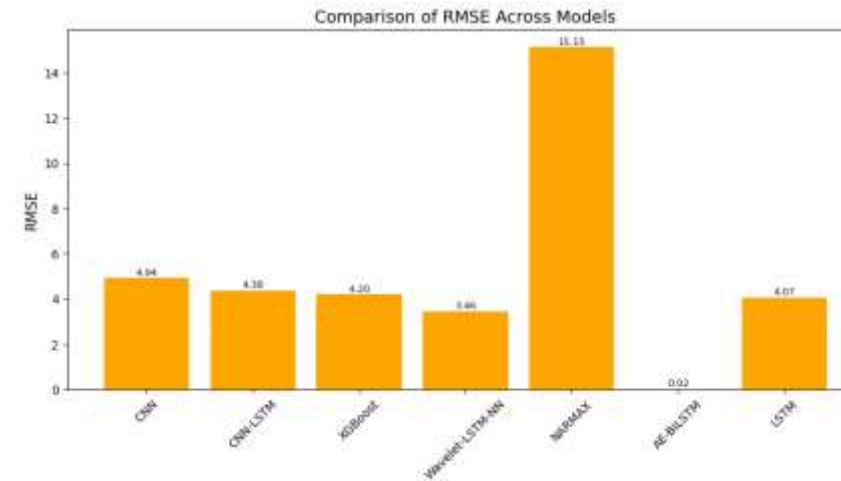
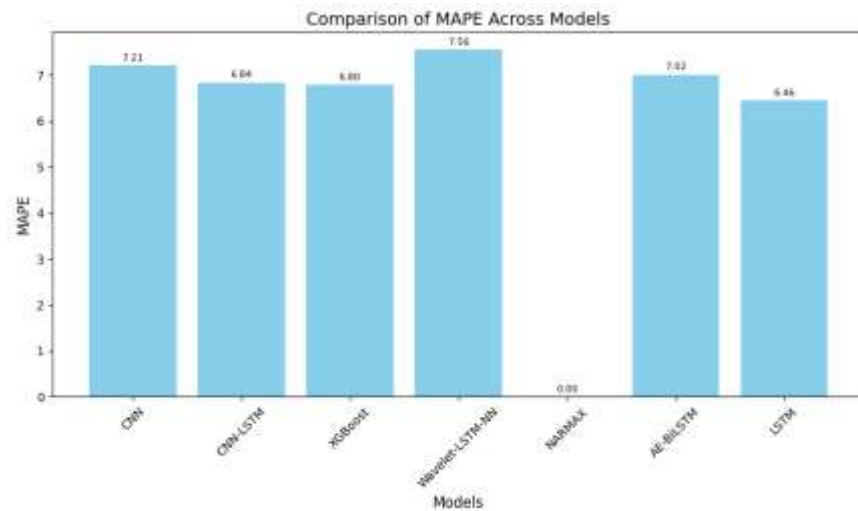
RMSE: 4.065

MAPE: 6.460



Model Comparison:

<u>Model</u>	<u>MAE</u>	<u>RMSE</u>	<u>MAPE</u>
CNN	1.591	4.939	7.211
CNN-LSTM	1.501	4.379	6.844
XGBoost	1.476	4.203	6.798
Wavelet-LSTM-NN	2.77	3.46	7.56
NARMAX	-	15.15	-
AE-BiLSTM	0.0057	0.0187	7.0189
LSTM	1.356	4.065	6.46



Possible Future Work:

- Although we are able to achieve good results with simple LSTM model , there can be hybrid model that can perform better with our models preprocessing.
- Can add explainable AI techniques to interpret the model's predictions, making the results more transparent and knowing impact of each step.

References

- [1] C. McHugh, S. Coleman, and D. Kerr, “Hourly electricity price forecasting with narimax,” Machine Learning with Applications, vol. 9, p. 100383, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S266682702200069X>
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Questions?



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Thank You
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